

Business Rule Testing (<i>not testing CRUD here, see next section</i>)

EP TCIs

- **name**
- **gpa**

TCI	Parameter	Range of Inputs
EP1	gpa	0..4.0
EP2	name	1..255

EP Test Cases (I am not covering values outside of the 2 EPs, as they will be caught in BVA testing)

Test Case ID	TCI Covered	gpa	name	Exp.
TC1.1	EP1[2]	2.5	Jim Hopper	Accepted as valid

BV TCIs

- **name**
- **gpa**

TCI	Parameter	Boundary Value
BV1*	gpa	-1.0
BV2		0
BV3		1.0
BV4		2.0
BV5		3.99

BV6		4.0
BV7*		4.1
BV8*	name	NULL
BV9		1
BV10		2
BV11		254
BV12		255
BV13*		256

BV Test Cases (grouping test cases together that yield the same results to optimize, [] = same outcome)

Test Case ID	TCI Covered	gpa	name	Exp.
BVA1.1*	BV1	-1.0	Jim Hopper	Exception: gpa outside of range
BVA1.2	BV2[3][4][5][6]	0.0	Jim Hopper	Accepted as valid
BVA1.3*	BV8[13]	3.5	Some name over 255 ch. OR no name :)	Exception: name outside of range
BVA1.4	BV9[10][11][12]	4.0	J	Accepted as valid
BVA1.5*	BV6	4.1	Jim Hopper	Exception: gpa outside range

DT Testing (not retesting cases from EP/BVA)

Cause

- Course Full?

Effect

- False = Can add student (rule 1)
- True = Cannot add student (rule 2)

Test Case ID	TCI Covered	gpa	name	Course Full?	Exp
DT1.1	DT1	3.5	Jim Hopper	F	Can add student
DT1.2	DT2	3.5	Jim Hopper	T	Cannot add student

CRUD Process Testing (*illustrated as state transitions*)

Student States

- Student does not exist yet
- Create student
- Update student info
- Delete a student

Student Transitions (like a TCI)

API Command	Current State	Resulting State
POST (create student)	DNE	Student created
PUT (update student)	Student created	Student info updated
DELETE (delete student)	Student created/updated	Student deleted
GET (get student)	Student created/updated	Student info read

Student Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST1.1	Student DNE	POST <code>"/{id}"</code>	Created	Student created
ST1.2	Student created	GET <code>"/{id}"</code>	Created	Student info returned
ST1.3	Student created	PUT <code>"/{id}"</code>	Updated	Student info updated
ST1.4	Student created	DELETE <code>"/{id}"</code>	Deleted	Student removed
ST1.5*	Student DNE	PUT <code>"/{id}"</code>	Student DNE	Error
ST1.6*	Student DNE	GET <code>"/{id}"</code>	Student DNE	Error

Course Transitions (like a TCI)

- Course does not exist yet
- Create course
- Update course info
- Delete a course

API Command	Current State	Resulting State
POST (create course)	DNE	Course created
PUT (update course)	Course created	Course info updated
DELETE (delete course)	Course created/updated	Course deleted

GET (get all courses)	Course created/updated	Course info returned
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Course Transitions (Test Case)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST2.1	Course DNE	POST /courses/{id }	Created	Course created
ST2.2	Course(s) created	PUT /courses/{id }	Updated	Course info updated
ST2.3	Course(s) created	DELETE /courses/{id }	Removed	Course deleted
ST2.4	Course(s) created	GET /courses/{id }	Retrieved	Course(s) retrieved
ST2.44*	Course DNE	PUT /courses/{id }	Course DNE	Error
ST2.45*	Course DNE	GET /courses/{id }	Course DNE	Error

Student + Course State Transitions (like a TCI)

- Add student to an existing course with an empty roster
- Add student to an existing course with students
- Delete student from an existing course with only that student
- Delete student from an existing course with many students

API Command	Current State	Resulting State
POST (add student to a course)	Empty Course	Has Student(s)
POST (add student to a course)	Has student(s)	Full
DELETE (remove student from course)	Has student	Empty
DELETE (remove student from course)	Full	Has student(s)

Student + Course State Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST2.5	Empty course	POST <code>/ {courseId} / students / {studentId}</code>	Has student(s)	Valid operation
ST2.6	Has student(s)	POST <code>/ {courseId} / students / {studentId}</code>	Full	Valid operation
ST2.7*	Full course	POST <code>/ {courseId} / students / {studentId}</code>	Full course	Invalid operation
ST2.8	Has student(s)	DELETE <code>/ {courseId} / students / {studentId}</code>	Empty/Has student(s)	Valid operation